DFT-informed stacked ensembles for experimental composition–property prediction: two pre-registered, fully nested evaluations on Matbench

Bhaskar Gurram

Independent Researcher,

Abstract

Experimental composition–property datasets are small, while computed properties of related compounds are abundant. We evaluate DFT-informed, cross-fitted stacked ensembles on the four composition tasks of Matbench v0.1: steel yield strength, experimental band gap, metallicity and glass formation. The protocol is strict: official folds only, every choice made by cross-validation inside each training fold, and the analysis plan and code frozen by SHA-256 hash before each scoring of the test folds.

Pipeline. Standard composition experts are stacked with simplex weights, with three additions:

- a disclosed transfer-learning block of PBE band gaps and formation energies from offline Materials Project data, matched by reduced formula;
- hurdle experts for the zero-inflated band gap;
- Optuna-tuned gradient boosting.

Results (paired over the same test compositions, using the leaders' public predictions):

- **band gap:** MAE 0.2352 eV against 0.2865 eV for the best leaderboard entry ($p_{\text{Holm}} < 0.001$);
- **metallicity and steel yield strength:** on par with the leaders;

Email address: gurrambhaskar.ai@gmail.com (Bhaskar Gurram)

- **glass formation:** clearly behind.

The leaders use no external data, so beating them on the band gap is not a like-for-like comparison of learning algorithms.

Post-hoc ablation (not used for selection):

- without the DFT block, the band-gap MAE rises to 0.2724 eV, which removes 72% of the margin over the leader;
- without the hurdle experts, it rises to 0.2789 eV;
- either variant is statistically on par with the leader;
- without the DFT block, metallicity falls significantly below the leader;
- steel and glass results do not change.

Negative finding. A novelty-damped local residual correction, tested in a first pre-registered run, transferred only weakly from our earlier chemistry work. It was never retained under nested validation.

Metric note. The leaderboard's classification "ROC-AUC" values for the leaders are in effect balanced accuracies.

All computation used one 24-core CPU workstation.

Keywords: materials informatics, experimental band gap, DFT transfer learning, Matbench, stacking, zero-inflated regression, nested cross-validation, pre-registration

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1. Introduction

Experimental materials properties are expensive to measure, so the datasets are small. Yet for many compounds a cheap computed estimate of a closely related property exists. The experimental band gap is the canonical example. Density-functional theory (DFT) with semi-local functionals underestimates gaps systematically, but the computed gap still carries much of the chemical trend. Large DFT databases such as the Materials Project (Jain et al., 2013) cover tens of thousands of compositions. Using computed values as inputs or as a low-fidelity source has a long history. Lee et al. (2016) used Kohn–Sham gaps (PBE and mBJ) together with elemental and compound predictors to predict G_0W_0 gaps of 270 inorganic compounds. The test RMSE fell from

0.59 eV, for a linear model on the Kohn–Sham gap alone, to 0.24 eV for support-vector regression on the full predictor set. Chen et al. (2021) trained multi-fidelity graph networks on PBE and higher-fidelity gaps. Including the low-fidelity PBE gaps reduced the MAE of experimental band-gap prediction by 22–45%. Their model, however, needs a crystal structure for every compound. Related approaches include multi-fidelity co-kriging (Pilania et al., 2017), stacked transfer from DFT-trained models (Kauwe et al., 2020), deep transfer learning (Jha et al., 2019), multi-fidelity denoising (Liu et al., 2022) and knowledge transfer for experimental gaps (Ko et al., 2024). Our DFT block is the simplest member of this family: a lookup of PBE quantities by reduced formula, used as features. It needs no structure for the target compound, which the composition-only Matbench tasks do not provide. On the public Matbench v0.1 benchmark (Dunn et al., 2020), however, the composition-only experimental tasks are led by models that use no external computed data. These are a fine-tuned scientific language model (Darwin) on the two band-gap-derived tasks, an AutoML pipeline (TPOT-Mat) on steels, and MODNet (De Breuck et al., 2021) on glass-forming ability (Matbench developers, 2026).

This paper asks how far a DFT-informed but otherwise conventional pipeline gets on these four tasks under a strict protocol. The pipeline is a cross-fitted stack of composition experts (Wolpert, 1992; Breiman, 1996; van der Laan et al., 2007). The protocol has four rules:

- only the official folds are used;
- every model, hyperparameter and design choice is made by cross-validation inside each training fold;
- the analysis plan is written, and the code frozen by SHA-256 hash, before any test fold is scored;
- every task, fold and comparison is reported.

We report two pre-registered runs.

- **First run.** It tested the transfer of a novelty-damped local out-of-fold residual correction from our earlier chemistry study (Gurram, 2026). The correction transferred only weakly, and we report this as a short negative finding.

- **Second run.** Written after the first run's results were seen, and disclosed as such, it added a DFT feature block, a hurdle model for the zero-inflated band gap (cf. the mixed-variable view of Ma et al., 2025), further descriptors and tuned gradient boosting.

Against the leaders, the DFT-informed pipeline:

- lowers the experimental band-gap MAE below the current best entry;
- is on par for metallicity and steel yield strength;
- remains clearly behind on glass formation.

The leaders used no external data, so beating them on expt_gap is not a like-for-like comparison of learning algorithms. It shows what disclosed transfer of computed data adds under a fully nested protocol. A post-hoc ablation, labelled as such, quantifies how much of the gain is due to the DFT features (Section 3.2):

- without them, 72% of the band-gap margin over the leader disappears, and the pipeline is on par with the leader rather than ahead of it;
- without them, the metallicity result falls significantly below the leader.

2. Methods

2.1. Tasks, splits and protocol

We use the four composition-only tasks of Matbench v0.1 with their official five folds, through `matbench`:

- steels: yield strength, 312 alloys, MAE;
- expt_gap: experimental band gap, 4,604 compounds, MAE;
- expt_is_metal: metallicity, 4,921 compounds, classification;
- glass: glass-forming ability, 5,680 alloys, classification.

Inside each official training fold T_k the rows are split into $J = 5$ inner folds (seed 0). All selection uses only T_k: hyperparameters, the choice among

design configurations, stack weights, the optional local correction and classification thresholds. The design space was explored on the training rows of fold 0 only.

Two analysis plans were frozen in turn, each by SHA-256 hashes of all code, and each run scored the test folds once.

- **Run 1 (Amendment 1).** Originally the design was fixed on fold 0's training rows. Amendment 1 moved the choice among the explored configurations inside every training fold. An earlier, superseded run under the original plan was never scored.
- **Run 2 (Amendment 2).** Written after run 1 had been scored.

Plans, amendments, freeze hashes, per-fold predictions and the post-hoc ablation (plan Section 12) are in the repository. The Supplementary Material (SI) gives details.

2.2. Features

All descriptors are deterministic functions of the formula and are computed with `matminer` (Ward et al., 2018a) and `pymatgen` (Ong et al., 2013):

- Magpie element-property statistics (Ward et al., 2016) and element fractions;
- stoichiometric, valence-orbital (Meredig et al., 2014), ionic, band-centre and atomic-orbital descriptors;
- the "deml", "matminer" and "megnet_el" element-property presets;
- Miedema enthalpies, Yang solid-solution parameters (Yang and Zhang, 2012) and atomic packing efficiency (Laws et al., 2015), as implemented by Ward et al. (2018b);
- Wen alloy descriptors and the transition-metal fraction;
- Oliynyk element statistics (Oliynyk et al., 2016).

DFT block (disclosed transfer learning). Each composition is matched by reduced formula to the Materials Project PBE data distributed with `matminer` (`matbench_mp_gap`, joined to `matbench_mp_e_form`). The block contains:

- the gap, metallicity and formation energy of the lowest-formation-energy polymorph;
- the minimum, median and maximum gap over polymorphs, the metallic fraction and the polymorph count;
- a match indicator.

It matches 73% of the expt_gap and expt_is_metal compositions, 1% of glass and none of steels. The block uses no experimental labels and no pretrained model.

2.3. Experts

- **Gradient boosting.** LightGBM (Ke et al., 2017) on all features, tuned inside T_k by Optuna (Akiba et al., 2019): 20 trials, with the default configuration evaluated first. Also XGBoost (Chen and Guestrin, 2016) and CatBoost (Prokhorenkova et al., 2018) with fixed settings.
- **Tree ensembles.** Extremely randomised trees (Geurts et al., 2006) on two feature sets and on element fractions alone.
- **Kernel, neighbour and neural experts** (Pedregosa et al., 2011): kernel ridge regression or RBF support-vector classifiers on element fractions and on Magpie features, distance-weighted k-nearest neighbours, and a two-layer perceptron.
- **Hurdle experts (expt_gap only).** About half of the experimental gaps are exactly zero. Each hurdle expert pairs a classifier for $E_g > 0$ with a regressor fitted on the $E_g > 0$ rows, and combines them as $p\,r$ or as $r \mathbb{1}[p \geq 0.5]$. The latter is the MAE-optimal rule for a point mass at zero plus a positive part.

CrabNet- (Wang et al., 2021) and MODNet-based experts were implemented but dropped, because a single fit exceeded our CPU budget.

2.4. Stack, gated correction and thresholds

Every expert is cross-fitted inside T_k to give out-of-fold (OOF) predictions. A simplex-constrained stack (non-negative weights summing to one) is fitted on these OOF predictions. Its loss is the MAE for regression and the Brier score for classification.

The local OOF residual correction of run 1 adds a kernel-weighted, novelty-damped average of neighbouring training residuals (SI Section S2). In run 2 it is kept only if a nested inner-CV check confirms that it improves the honest OOF score.

Hard labels use the threshold that maximises balanced accuracy on the training OOF predictions.

2.5. Endpoints and statistics

The endpoints follow the board's semantics.

- **Regression:** MAE.
- **Classification:** balanced accuracy, because the leaders either recorded labels (Darwin) or were scored by a `matbench` version that thresholded probabilities at 0.5 (MODNet v0.1.12), which makes their reported "ROC-AUC" equal to their balanced accuracy.
- **Run 2 primary comparison:** our final pipeline versus each task's leader, paired over the same test compositions using the leaders' public per-sample predictions.

The test is a bootstrap over test points within folds ($B = 10^4$), with Holm correction over the four tasks. We claim to beat a leader only if our mean is better and the Holm-adjusted $p < 0.05$. Run 1's primary endpoint (corrected versus uncorrected stack) used the same bootstrap. The same statistics are used for the post-hoc ablation, which is descriptive only.

3. Results

3.1. The DFT-informed pipeline versus the leaderboard (run 2)

Table 1 and Fig. 1 compare the frozen run-2 pipeline with each task's leader.

Table 1: Run-2 (Amendment-2) final pipeline versus each task's leaderboard leader, paired over the official test compositions using the leaders' public per-sample predictions. Metrics follow the board: MAE; balanced accuracy for classification. Δ is ours minus the leader. 95% CI: paired bootstrap over test points within folds. Holm correction over four tasks. Rank is among the entries of 18 October 2026, including ours.

Task	Metric	Final pipeline	Leader (board)	Δ [95% CI]	p_{Holm}	Rank
steels	MAE	80.10 ± 10.19	TPOT-Mat: 79.95	+0.1577 [-4.54, +4.89]	0.928	2 of 12
expt_gap	MAE	0.2352 ± 0.0167	Darwin: 0.2865	-0.05131 [-0.0701, -0.0331]	<0.001	1 of 13
expt_is_metal	bal. acc.	0.9547 ± 0.0033	Darwin: 0.9598	-0.005093 [-0.0118, +0.00179]	0.299	2 of 8
glass	bal. acc.	0.8937 ± 0.0158	MODNet (v0.1.12): 0.9603	-0.06658 [-0.0759, -0.0573]	<0.001	2 of 8

expt_ gap. The MAE is 0.2352 eV against 0.2865 eV for Darwin. The difference is $\Delta = -0.05131$ eV (95% CI [-0.0701, -0.0331]; $p_{\text{Holm}} < 0.001$). This is the only task that meets our pre-registered rule for beating the leader.

expt_ is_ metal and steels. The pipeline is statistically on par with the leaders:

- expt_is_metal: balanced accuracy 0.9547 versus 0.9598 ($p_{\text{Holm}} = 0.299$);
- steels: MAE 80.10 versus 79.95 MPa ($p_{\text{Holm}} = 0.928$).

glass. The pipeline is clearly behind MODNet v0.1.12: 0.8937 versus 0.9603 ($p_{\text{Holm}} < 0.001$). This also holds for the probability ROC-AUC (0.9606 versus 0.9899).

Caveat. These are not like-for-like comparisons. Our pipeline uses an external DFT block for expt_gap and expt_is_metal, and the leaders did not (Section 4).

3.2. How much of the gain is due to DFT features: post-hoc ablation

The ablation was written after run 2 had been scored and was not used for any selection. The reported pipeline and the prepared leaderboard files are unchanged. It re-runs the frozen run-2 code with one component removed:

- without the DFT block, on all four tasks;
- without the three hurdle experts, on expt_gap only.

Each variant was scored once, using the run-2 paired bootstrap (Table 2). The p-values are descriptive and unadjusted.

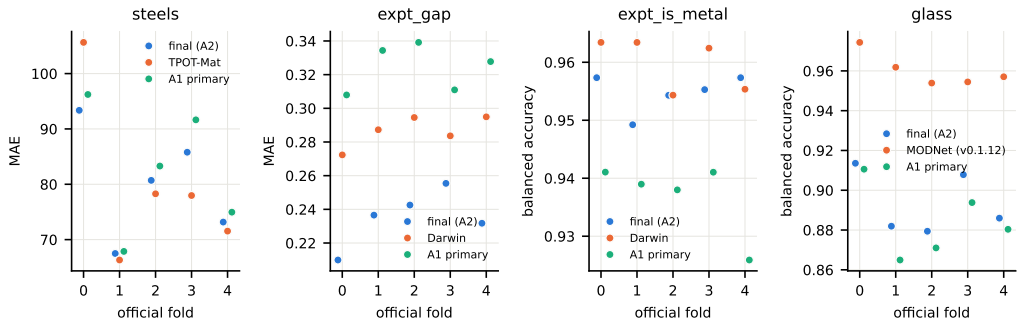


Figure 1: Per-fold board metric of the run-2 final pipeline, the task leader and the run-1 primary arm.

expt_ gap.

- **Without the DFT block,** the MAE rises from 0.2352 to 0.2724 eV ($\Delta = +0.0372$ eV, 95% CI [+0.0251, +0.0493], +15.8%, $p_{\text{boot}} < 0.001$). It is worse in 5/5 folds. This removes 72% of the full pipeline's -0.05131 eV margin over Darwin.
- **Without the hurdle experts,** the MAE rises to 0.2789 eV ($\Delta =$ +0.0437 eV [+0.0351, +0.0520], +18.6%, $p_{\text{boot}} < 0.001$; 5/5 folds). This removes 85% of the margin.
- **The two components are not additive.** Each removal alone erases most of the margin. Each variant is statistically on par with Darwin:
  - without DFT: $\Delta = -0.0141$ eV, $p_{\text{boot}} = 0.129$;
  - without hurdles: $\Delta = -0.0077$ eV, $p_{\text{boot}} = 0.414$.

  Under our claim rule, therefore, neither the DFT block nor the hurdle experts alone would have been enough to beat the leader. This is consistent with the two carrying overlapping information, because the DFT gap and metal flag also indicate which compounds have $E_g = 0$.

expt_ is_ metal. Without the DFT block the balanced accuracy falls from 0.9547 to 0.9425 ($\Delta = -0.0122$ [-0.0177, -0.0069], $p_{\text{boot}} < 0.001$; 5/5 folds).

- The variant is significantly below Darwin ($\Delta = -0.0173$, $p_{\text{boot}} < 0.001$), so parity with the leader on this task also depends on the DFT block.
- The probability ROC-AUC falls less, from 0.9895 to 0.9849.

Table 2: Post-hoc ablation of the frozen run-2 (A2) pipeline (plan Section 12; written after run-2 scoring and not used for selection). Board metric, as mean ± s.d. over the five official folds, of the full A2 pipeline and of each variant. Δ: variant minus full A2, and variant minus the task leader, with the 95% paired-bootstrap CI ($B = 10^4$, resampling test points within folds) and unadjusted p_{boot}. "Folds worse": folds in which the variant is worse than the full pipeline. MAE: lower is better. Balanced accuracy: higher is better. Generated from `results/ablation_summary.json`.

		mean ± s.d.		variant − full A2		variant − leader		
Task	Variant	Full A2	Variant	Δ [95% CI]	p_{boot} (folds worse)	Leader	Δ [95% CI]	p_{boot}
expt_gap MAE (eV)	no DFT block	0.2352 ±0.0167	0.2724 ±0.0088	+0.0372 [+0.0251, +0.0493]	<0.001 (5/5)	0.2865 Darwin	-0.0141 [-0.0326, +0.0043]	0.129
expt_gap MAE (eV)	no hurdle experts	0.2352 ±0.0167	0.2789 ±0.0107	+0.0437 [+0.0351, +0.0520]	<0.001 (5/5)	0.2865 Darwin	-0.0077 [-0.0256, +0.0108]	0.414
expt_is_metal bal. acc.	no DFT block	0.9547 ±0.0033	0.9425 ±0.0050	-0.0122 [-0.0177, -0.0069]	<0.001 (5/5)	0.9598 Darwin	-0.0173 [-0.0244, -0.0102]	<0.001
steels MAE (MPa)	no DFT block	80.10 ±10.19	80.06 ±10.28	-0.04 [-0.54, +0.46]	0.877 (1/5)	79.95 TPOT-Mat	+0.11 [-4.80, +4.88]	0.978
glass bal. acc.	no DFT block	0.8937 ±0.0158	0.8922 ±0.0141	-0.0016 [-0.0054, +0.0021]	0.416 (3/5)	0.9603 MODNet (v0.1.12)	-0.0682 [-0.0777, -0.0588]	<0.001

steels and glass. The DFT block matches no steels and 1% of glass compositions, so it serves mainly as a negative control. Removing it changes nothing detectable:

- steels: 80.06 versus 80.10 MPa ($p_{\text{boot}} = 0.877$);
- glass: 0.8922 versus 0.8937 balanced accuracy ($p_{\text{boot}} = 0.416$).

The small per-fold differences (SI Section S8) most likely arise because removing these constant or nearly constant columns changes the random column subsampling of the tree ensembles.

3.3. The local residual correction: a negative finding (run 1)

Run 1 pre-registered a single primary question: does the novelty-damped local OOF residual correction improve a cross-fitted stack on these tasks?

- **expt_gap.** It improved the MAE from 0.3264 to 0.3240 eV (-0.71%; 5/5 folds; $p_{\text{Holm}} = 0.005$).
- **steels.** No detectable effect (-0.39%).
- **Classification tasks.** The inner selection almost never switched the correction on (0/5 and 1/5 folds).

Table 3: Board metric of the two pre-registered pipelines on the official folds, with a paired bootstrap p-value for the difference.

Task	Amendment 1 (primary arm)	Amendment 2 (final)	p_{boot}
steels	82.80	80.10	0.288
expt_gap	0.3240	0.2352	<0.001
expt_is_metal	0.9370	0.9547	<0.001
glass	0.8841	0.8937	<0.001

Stratifying the test errors by distance to the nearest training composition showed the correction *hurting* for near-duplicate compositions (expt_gap +0.9%, steels +6.8% in the nearest quartile). A fold-matched variant designed to remove the double counting behind this did not help (+1.10% MAE versus the naive correction on expt_gap). In run 2 the nested gate kept the correction in none of the 20 folds. The SI reports all run-1 arms, ablations and controls.

3.4. Stacking and the two runs

Run 2 improved on run 1 on every task (Table 3). The improvement is large on expt_gap and expt_is_metal, and within noise on steels ($p = 0.288$). The simplex stack beat the best single expert selected by inner CV only on glass (+0.18% ROC-AUC, $p = 0.014$). Elsewhere it matched that expert. On steels, extremely randomised trees on element fractions alone reached 79.56 MPa as the best single expert, against 80.10 MPa for the pre-registered stack, which we report. One official fold took a median of 9–62 minutes on a 24-core CPU workstation with five parallel workers.

4. Discussion

Like-for-like caveat, quantified. The expt_gap result is not a like-for-like comparison of learning algorithms.

- **The leaders used no external data.** As model inputs they used only the chemical formula and the task's own training labels. TPOT-Mat and MODNet use composition descriptors. Darwin is a language model pre-trained on scientific text and fine-tuned on the task.
- **Our pipeline does use external data.** It feeds PBE band gaps and formation energies from the Materials Project into the experts.

The ablation (Section 3.2) puts a number on this.

- Removing the DFT block raises our band-gap MAE by +0.0372 eV (+15.8%), which is 72% of our margin over Darwin.
- The remaining DFT-free pipeline is statistically on par with Darwin (0.2724 versus 0.2865 eV, $p_{\text{boot}} = 0.129$), not ahead of it.
- On expt_is_metal the DFT block likewise turns a result significantly below Darwin into parity.

The claim that survives is therefore:

- a DFT-free, fully nested stack is on par with the leader on expt_gap, but below it on expt_is_metal;
- disclosed DFT transfer adds a further, clearly significant improvement;
- that improvement, together with the hurdle experts, takes the pipeline past the leader on expt_gap and to parity on expt_is_metal.

The hurdle experts matter to a similar degree: without them, too, most of the margin disappears.

Relation to prior DFT-informed gap models. That computed gaps are informative inputs or low-fidelity sources for experimental gaps is well established (Lee et al., 2016; Pilania et al., 2017; Kauwe et al., 2020; Chen et al., 2021; Liu et al., 2022; Ko et al., 2024).

- **Lee et al. (2016)** showed, for 270 compounds and G_0W_0 targets, that combining Kohn–Sham gaps with elemental predictors in a nonlinear model reduces the RMSE from 0.59 to 0.24 eV, compared with a linear correction of the Kohn–Sham gap alone.
- **Chen et al. (2021)** reported a 22–45% lower experimental-gap MAE when low-fidelity PBE gaps were added to multi-fidelity graph networks.

Adding our DFT block lowers the MAE by 13.6% relative to the DFT-free variant. This is smaller than the improvement Chen et al. (2021) report, and it is obtained in a different setting:

- from a formula lookup, without any structure for the target compound;
- on the official Matbench folds;
- with the DFT block covering only 73% of the compositions, the remainder being flagged as unmatched.

The results are not directly comparable, because the datasets, splits, targets and metrics differ.

Where the pipeline does not lead. On glass, MODNet v0.1.12's public predictions reach a probability ROC-AUC of 0.9899, against our 0.9606. Neither the glass-formation descriptors nor any CPU-feasible expert we tried closed this gap. On expt_is_metal and steels the pipeline is statistically on par with the leaders. On expt_is_metal this parity depends on the DFT block. We did not audit the leaders' protocols, for example whether all of their tuning was confined to the training folds.

Classification metrics. The board's "ROC-AUC" for the classification leaders is, in effect, balanced accuracy. Darwin recorded labels. MODNet recorded probabilities that the `matbench` version used for its board entry thresholded at 0.5. Our probability ROC-AUC on expt_is_metal (0.9895) is therefore not comparable to the board's numbers. We recommend reporting both quantities in separate columns.

The local correction. The local residual correction that motivated this work transferred only weakly from chemistry to materials. Once its benefit had to survive a nested inner-CV check, it was never kept. We report it as a negative result.

Limitations.

- **Two looks at the test folds.** Run 2 was written after run 1 had been scored, so its motivation, though none of its choices, depended on the first test results. Both runs are reported.
- **Candidate space from fold 0.** The candidate feature blocks and expert pool were developed on fold 0's training rows. Those rows contain the test rows of folds 1–4.

- **Optimistic tuning OOF.** The tuned expert's OOF predictions are mildly optimistic, because tuning and stacking share inner splits. This can affect only the stack weights.

- **Scope.** Four tasks, CPU-feasible experts only, five folds and a single seed limit the power and generality of the conclusions.

- **Post-hoc ablation.** The ablation was designed after run-2 scoring, is descriptive and reports unadjusted p-values. Each component was removed one at a time, so the effects are not additive. It uses one seed, and does not separate the gap features from the formation-energy features within the DFT block.

5. Conclusions

Under two pre-registered, fully nested protocols on the Matbench v0.1 composition tasks, a cross-fitted stack of standard experts with a disclosed DFT feature block and a hurdle model for zero-inflated gaps:

- lowered the experimental band-gap MAE to 0.2352 eV, below the current leader, which used no external data; without the DFT block the MAE was 0.2724 eV, on par with that leader;

- was on par with the leaders for metallicity (only with the DFT block) and steel yield strength;

- remained behind on glass formation.

The local residual correction that originally motivated the study did not survive nested validation.

Data and code availability

All code, the frozen analysis plan (`ANALYSIS_PLAN.md`), SHA-256 freeze hashes, per-fold predictions of every arm, per-fold metrics and the unsubmitted Matbench result files are in the repository `https://github.com/bhaskargurram-ai/matbench-composition-stack`, which will be made public on submission. The data are the public Matbench v0.1 tasks.

Affiliation

The author is an independent researcher with no institutional affiliation.

CRediT authorship contribution statement

Bhaskar Gurram: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic), running the ResearchStudio idea-generation skills, to assist with literature search, prior-art checking, code development and drafting of the manuscript. All analysis code, results and text were reviewed by the author, who takes full responsibility for the content of the publication.

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